Exercise 7: NumPy practice

1. Write code which can create a 2D NumPy array of a given size, which is filled with zeros and has ones all around its border (i.e. left and right columns, and top and bottom rows). Is there a very compact way to achieve this?

2. Noughts and Crosses: create a Python program which:

(a) Creates an empty 3x3 NumPy array of a suitable datatype;

(b) Accepts input from a player, allowing them to specify a row and column in the matrix (array);

(c) Checks to see whether there is already a nought or a cross in that cell, and, if not, adds one;

(d) Checks to see whether the game has been won (that is, there are either three crosses or three noughts in any diagonal, horizontal or vertical row);

(e) If the game has not been won, repeats from part (b) so that the player can enter another cell.

Optionally, you could choose to print out the NumPy matrix before each turn (so that the players can see the state of the playing board), and to keep track of whose turn it is (printing out "Noughts" or "Crosses" each time input is being requested, depending on which token is being played).

3. Write code which creates a 3x3x3 array where the first element is 0, the last is 6.5, and the elements in between are all linearly-spaced, increasing values.

(a) Which index of your array contains the value "5"?

(b) Using your array, and any NumPy methods you believe are useful, identify which index of a _1D version of that array_ would contain the value "5".

(c) Write standard Python looping statements to iterate through the array, and verify that the index you have from (a) actually refers to the correct element.

(d) Optionally, adapt your code (or write new code) which will allow you to perform the above operations on _any_ dimension of 3D array, and for any sought value (have your code gracefully handle the situation where that value is not present in the array).